

$$1) a) \quad y[n] = x[n] - x[n-1]$$

$$h[n] = \delta[n] - \delta[n-1]$$

$$\text{or } h[n] = \{1, -1\}$$

$$b) \quad y[0] = x[0] - x[-1] = 1 - 0 = 1$$

$$y[1] = x[1] - x[0] = 1 - 1 = 0$$

$$y[2] = x[2] - x[1] = 0 - 1 = -1$$

$$y[3] = x[3] - x[2] = 0 - 0 = 0$$

$$\therefore y[n] = \{1, 0, -1\}$$

$$c) \quad H(z) = 1 - z^{-1}$$

$$d) \quad X(z) = 1 + z^{-1}$$

$$e) \quad Y(z) = H(z)X(z)$$

$$Y(z) = (1 - z^{-1})(1 + z^{-1})$$

$$Y(z) = 1 - z^{-2}$$

$$f) \quad y[n] = \delta[n] - \delta[n-2]$$

$$y[n] = \{1, 0, -1\}$$

i.e. Same as part (b).

$$2) \quad y[n] = x[n] + 2x[n-1] + x[n-2]$$

$$x[n] = \{1, -1\}$$

$$a) \quad h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$$

$$h[n] = \{1, 2, 1\}$$

$$b) \quad y[0] = x[0] + 2x[-1] + x[-2] = 1 + 0 + 0 = 1$$

$$y[1] = x[1] + 2x[0] + x[-1] = -1 + 2 + 0 = 1$$

$$y[2] = x[2] + 2x[1] + x[0] = 0 - 2 + 1 = -1$$

$$y[3] = x[3] + 2x[2] + x[1] = 0 + 0 - 1 = -1$$

$$y[n] = \{1, 1, -1, -1\}$$

$$c) \quad H(z) = 1 + 2z^{-1} + z^{-2}$$

$$d) \quad X(z) = 1 - z^{-1}$$

$$e) \quad Y(z) = X(z)H(z)$$

$$Y(z) = (1 - z^{-1})(1 + 2z^{-1} + z^{-2})$$

$$Y(z) = 1 + 2z^{-1} + z^{-2} - z^{-1} - 2z^{-2} - z^{-3}$$

$$Y(z) = 1 + z^{-1} - z^{-2} - z^{-3}$$

$$f) \quad y[n] = \delta[n] + \delta[n-1] - \delta[n-2] - \delta[n-3]$$

$$y[n] = \{1, 1, -1, -1\}$$

$\therefore$ i.e. same as part (b).

$$3) y[n] = 0.8y[n-1] - 0.2x[n]$$

$$a) x[n] = \delta[n] = 1$$

$$y[0] = 0.8y[-1] - 0.2x[0] = 0 - 0.2 = -0.2$$

$$y[1] = 0.8y[0] - 0.2x[1] = 0.8 \times -0.2 - 0 = -0.16$$

$$y[2] = 0.8y[1] - 0.2x[2] = 0.8 \times -0.16 - 0 = -0.128$$

$$y[3] = 0.8y[2] - 0.2x[3] = 0.8 \times -0.128 - 0 = -0.1024$$

$$y[n] = \{-0.2, -0.16, -0.128, -0.1024\}$$

$$b) y[n] = -0.2 \times 0.8^n$$

$$y[100] = -0.2 \times 0.8^{100} = -4.07 \times 10^{-11}$$

c) Impulse response $x[n] = \delta[n]$
Already solved in (a) and (b)

$$\text{so } h[n] = -0.2 \times 0.8^n \quad n \geq 0$$

$$d) Y(z) = 0.8Y(z)z^{-1} - 0.2X(z)$$

$$Y(z) - 0.8Y(z)z^{-1} = -0.2X(z)$$

$$Y(z)(1 - 0.8z^{-1}) = -0.2X(z)$$

$$Y(z) = \frac{-0.2X(z)}{(1 - 0.8z^{-1})}$$

$$e) H(z) = \frac{Y(z)}{X(z)} = \frac{-0.2}{(1 - 0.8z^{-1})}$$

$$f) a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}} \quad \text{here } a = 0.8$$

$$\therefore h[n] = -0.2 \times 0.8^n u[n]$$

i.e. same as (b).

4) a) $y[n] = 0.75x[n] - 0.25x[n-1]$
 $h[n] = 0.75\delta[n] - 0.25\delta[n-1]$
 $h[n] = \{0.75, -0.25\}$ FIR filter

b) $y[n] = 0.75y[n-1] - 0.25x[n]$
 $Y(z) = 0.75Y(z)z^{-1} - 0.25X(z)$

$Y(z) - 0.75Y(z)z^{-1} = -0.25X(z)$
 $Y(z)(1 - 0.75z^{-1}) = -0.25X(z)$

$Y(z) = \frac{-0.25X(z)}{1 - 0.75z^{-1}}$

$H(z) = \frac{Y(z)}{X(z)} = \frac{-0.25}{1 - 0.75z^{-1}}$

$h[n] = z^{-1} \{H(z)\}$

$a^n u[n] \longleftrightarrow \frac{1}{1 - az^{-1}} \quad a=0.75$

$h[n] = -0.25 \times 0.75^n u[n]$

IIR filter

$$5) a) \{1, 2, 3\} = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

$$b) \{1, 2, 3\} = \delta[n+1] + 2\delta[n] + 3\delta[n-1]$$

$$c) 1\delta[n] - 3\delta[n-2] + 5\delta[n-4] \\ = \{1, 0, -3, 0, 5\}$$

$$d) y[n] = \sum_{m=0}^2 h[m]x[n-m] \\ x[n] = \{1, 1, 1\} \quad h[n] = \{1, 1, 1\}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$y[n] = \{1, 2, 3, 2, 1\}$$

$$e) \begin{array}{ccc} 3 & 2 & 1 \\ & 1 & 1 \end{array} \quad \begin{array}{ccc} 3 & 2 & 1 \\ & 1 & 1 \end{array} \quad \begin{array}{ccc} 3 & 2 & 1 \\ & 1 & 1 \end{array} \quad \begin{array}{ccc} 3 & 2 & 1 \\ & 1 & 1 \end{array}$$

$$y[n] = \{1, 3, 5, 3\}$$

$$f) x[n] = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$\text{extent } \{x[n]\} = 10$$

$$h[n] = \{8, 7, 6, 5, 4, 3, 2, 1\}$$

$$\text{extent } \{h[n]\} = 8$$

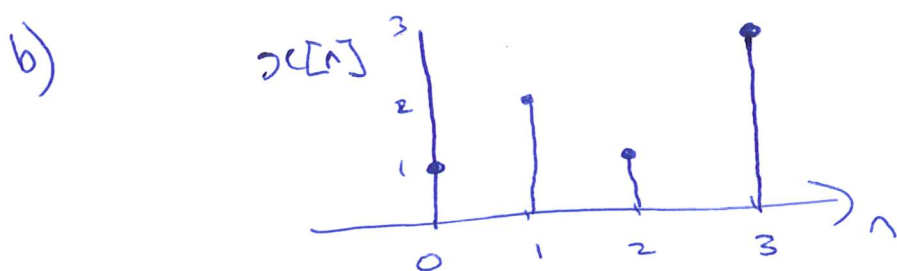
$$\text{extent } \{y[n] = h[n] * x[n]\} = 10 + 8 - 1 \\ = 17$$

5) g) Ignoring multiplications by zero
 $N = \text{Extent}\{x[n]\} \times \text{Extent}\{h[n]\}$
 $N = 10 \times 8 = 80$

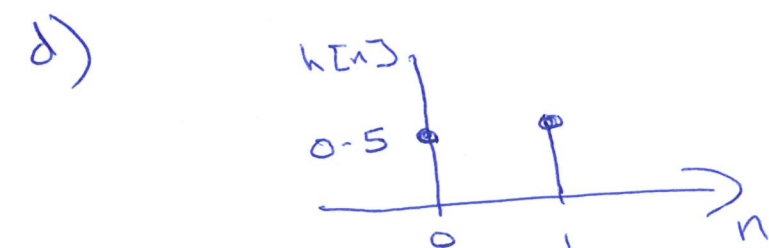
h) $\Rightarrow 100 + 50 - 1 = 149$

i) $N = 100 \times 50 = 5000$

6) $x[n] = \{1, 2, 1, 3\}$ $h[n] = \{0.5, 0.5\}$
 a) $x[n] = \delta[n] + 2\delta[n-1] + \delta[n-2] + 3\delta[n-3]$



c) $h[n] = \{0.5, 0.5\}$
 $h[n] = 0.5\delta[n] + 0.5\delta[n-1]$



$$e) \quad y[n] = \sum_{m=0}^1 h[m] x[n-m]$$

$$y[n] = h[0] x[n] + h[1] x[n-1]$$

$$y[0] = 0.5 \times 1 + 0.5 \times 0 = 0.5$$

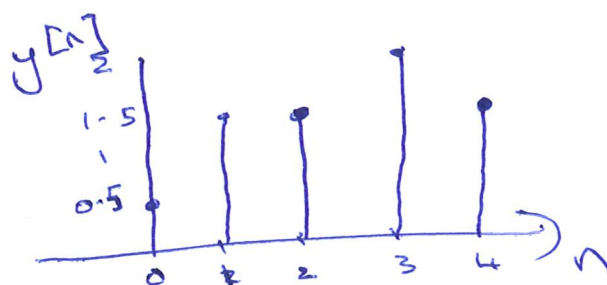
$$y[1] = 0.5 \times 2 + 0.5 \times 1 = 1.5$$

$$y[2] = 0.5 \times 1 + 0.5 \times 2 = 1.5$$

$$y[3] = 0.5 \times 3 + 0.5 \times 1 = 2$$

$$y[4] = 0.5 \times 0 + 0.5 \times 3 = 1.5$$

$$y[n] = \{0.5, 1.5, 1.5, 2, 1.5\}$$



$$f) \quad H(z) = 0.5 + 0.5 z^{-1} = 0.5(1 + z^{-1})$$

g) 2 Sample Moving Average.

$$7) a) \Delta f = \frac{f_s}{N} = \frac{1000}{1024} = 0.977 \text{ Hz}$$

$$b) \Delta f = \frac{f_s}{N} = \frac{1000}{4096} = 0.244 \text{ Hz}$$

$$c) N = 512, \Delta f = 1 \text{ kHz}$$

$$\Delta f = \frac{f_s}{N}$$

$$f_s = N \cdot \Delta f = 512 \times 1000 = 512000$$

$$T = \frac{1}{f_s} = \frac{1}{512000} = 1.95 \mu s$$

Sampling period

d) The purpose is to avoid circular convolution effects.

$$e) \text{Extent}\{h[n]\} = 100 \quad \text{Extent}\{x[n]\} = 150$$

$$\therefore \text{Extent}\{y[n]\} = 100 + 150 - 1 = 249$$

FFT is more efficient when N is a power of 2. Use $N = 256$ i.e. 2^8

7) f)

$$x[n] = \{1, 2, 2, 1\} \quad N=4$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}$$

$k=0$

$$X[0] = \sum_{n=0}^3 x[n] e^{-j0} = \sum_{n=0}^3 x[n] = 1+2+2+1 = 6$$

$k=1$

$$X[1] = \sum_{n=0}^3 x[n] e^{-j \frac{2\pi n}{4}} = 1e^{-j0} + 2e^{-j\frac{\pi}{2}} + 2e^{-j\pi} + 1e^{-j\frac{3\pi}{2}}$$

$$= 1 - 2j - 2 + 1j$$

$$= -1 - j$$

$k=2$

$$X[2] = \sum_{n=0}^3 x[n] e^{-j\pi n} = 1e^{-j0} + 2e^{-j\pi} + 2e^{-j2\pi} + 1e^{-j3\pi}$$

$$= 1 - 2 + 2 - 1$$

$$= 0$$

$k=3$

$$X[3] = \sum_{n=0}^3 x[n] e^{-j \frac{3\pi n}{2}} = 1e^{-j0} + 2e^{-j\frac{3\pi}{2}} + 2e^{-j3\pi} + 1e^{-j\frac{9\pi}{2}}$$

$$= 1 + 2j - 2 - j$$

$$= -1 + j$$

$$X[k] = \{6, -1-j, 0, -1+j\}$$

g) A window function is applied before using a FFT to reduce spectral leakage caused by truncating a signal. This truncation is equivalent to multiplying by a rect function in the time domain; or convolving by a sinc function in the frequency domain.